Supplementary File S3. Verification report for VerifyMyMU version 1.1.0

Nine top-down measurement uncertainty formulations on one clinical chemistry dataset: how much of a reported uncertainty depends on choices the guidance leaves open — Muhammed Fevzi Kilinckaya, Scandinavian Journal of Clinical and Laboratory Investigation.

This file documents the verification described in the Methods section ("Software verification and validation", five levels). For every level it gives what was checked, how, and the verbatim output of the check script. The scripts, the application and the example data are in the Zenodo record (doi:10.5281/zenodo.23142868) and the repository https://github.com/mfkilinckaya-svg/VerifyMyMU, so every output below can be regenerated by a reader.

1. Object under verification

Item	Value
File	VerifyMyMU.html, version 1.1.0 (the file supplied as Supplementary File S5 and archived at Zenodo under the same concept record as version 1.0.0, doi:10.5281/zenodo.23142868). Version 1.1.0 differs from 1.0.0, the version used for the analyses in the paper, only in the version string and in the wording of the applicability-screen label (“sensitivity only” → “agreement only”); the statement-level difference between the two files is reproduced in section 1.1 below.
MD5	b44fa0b4e52d6b3e311bbca219884702 (version 1.0.0: bd0d9ec4887eeb4e5768d66d0825900a)
SHA-256	a0c16c55e5664467b172016e133b608f34100a276add18148c1e171ecf535052 (version 1.0.0: e0dcd561982f30bb5e24a95e31f0e17259d866b9181df7bf697c3a7484a19930)
Run environment for the outputs reproduced in this file	Linux x86-64; Python 3.13.17; numpy, scipy; Playwright 1.56 driving Chromium (Chrome for Testing 153.0.8010.12), headless; 10 November 2026. All browser-driven scripts were re-run on version 1.1.0 on that date; every output line reproduced below is identical to the 25 October 2026 run on version 1.0.0 (126 output lines compared verbatim)
How the browser-driven scripts work	Each script opens VerifyMyMU.html in a headless browser as a user would, calls the same functions the screens call, reads the values the screens show, and compares them with an independent computation written in Python without reference to the application's code.

1.1 Difference between version 1.1.0 and version 1.0.0

The two files were split at every semicolon and compared statement by statement (diff of the two files after tr ';' '\n'). Fourteen statements differ, listed below with the 1.0.0 form marked < and the 1.1.0 form marked >; each is a display string (the applicability-screen label, which the paper now calls "agreement only" at a reviewer's suggestion) or the version string. No computational statement differs, so the Level 3a grid of 4662 comparisons recorded on 28 September 2026 against version 1.0.0 (section 5) applies unchanged to version 1.1.0; the other levels were re-run on 1.1.0 and are reproduced below.

```
<         <div class="sub">ISO/TS 20914 · GUM · v1.0.0</div>
>         <div class="sub">ISO/TS 20914 · GUM · v1.1.0</div>
<         and are labelled <i>sensitivity</i>. This view is for method comparison
>         and are labelled <i>agreement only</i>. This view is for method comparison
```

```
< <script>const APP_VERSION="1.0.0"
> <script>const APP_VERSION="1.1.0"
<   if(a.ref==='peer') return {v:'sens',why:'<b>Sensitivity only.</b> C<sub>ref</sub> is a method-
group consensus, so what you obtain is the uncertain …
>   if(a.ref==='peer') return {v:'sens',why:'<b>Agreement only.</b> C<sub>ref</sub> is a method-group
consensus, so what you obtain is the uncertainty …
<       +({ok:'applicable',sens:'sensitivity',warn:'check first',block:'not applicable',no:'not
available'}[v.v])+'</text>'
>       +({ok:'applicable',sens:'agreement only',warn:'check first',block:'not applicable',no:'not
available'}[v.v])+'</text>'
<             sens:['⚠️','#92400e','#fffbeb','#fde68a','Sensitivity only'],
>             sens:['⚠️','#92400e','#fffbeb','#fde68a','Agreement only'],
< color:#78350f"><b>Applicability screen — peer-group EQA.</b> Every EQA-based model below is
<i>computable</i>, but with a consensus target the bia …
> color:#78350f"><b>Applicability screen — peer-group EQA.</b> Every EQA-based model below is
<i>computable</i>, but with a consensus target the bia …
<   if(peer)return {s:'sens',t:'Sensitivity only — peer-group agreement, not traceable trueness'}
>   if(peer)return {s:'sens',t:'Agreement only — peer-group agreement, not traceable trueness'}
< const _mlBadge=s=>({ok:['#065f46','#d1fae5','applicable'],sens:['#92400e','#fef3c7','sensitivity'],
> const _mlBadge=s=>({ok:['#065f46','#d1fae5','applicable'],sens:['#92400e','#fef3c7','agreement
only'],
<       +'<b>applicable</b>: the model answers a question this data can answer. <b>sensitivity</b>:
computable, but the EQA target is a peer-group con …
>       +'<b>applicable</b>: the model answers a question this data can answer. <b>agreement
only</b>: computable, but the EQA target is a peer-group  …
<       +'agreement with the peer group, not metrological trueness. Those rows are labelled
<i>sensitivity</i> and must not be read as traceable bias. …
>       +'agreement with the peer group, not metrological trueness. Those rows are labelled
<i>agreement only</i> and must not be read as traceable bi …
<       const stTxt={ok:'applicable',sens:'sensitivity',warn:'check',block:'not applicable'}[st.s]
>       const stTxt={ok:'applicable',sens:'agreement only',warn:'check',block:'not applicable'}[st.s]
<   L.push('# VerifyMyMU v1.0.0 - method comparison appendix')
>   L.push('# VerifyMyMU v1.1.0 - method comparison appendix')
<   const foot=`<div class="rfoot"><span class="fbrand">VerifyMyMU v1.0.0 · Measurement Uncertainty
Report</span> — Pooled u<sub>Rw</sub> = ${poolF …
>   const foot=`<div class="rfoot"><span class="fbrand">VerifyMyMU v1.1.0 · Measurement Uncertainty
Report</span> — Pooled u<sub>Rw</sub> = ${poolF …
```

2. Summary of the five levels

Level	What is checked	Script	Result
1 · Source audit	Every implemented equation read against the primary document and recorded with its clause or page	Supplementary Table S1; the tool's "Formulas & sources" screen	Nine formulations traced to ISO/TS 20914, Nordtest TR 537 ed. 4, EUROLAB TR 1/2007, SH GTA 14, Martinello et al. 2020 and ISO 21748
2 · Worked examples	Reproduction of the published worked examples to the printed digits	nordtest_val.py (ammonium example, §3.2 / Appendix 4); §6.1, §6.2 and ISO/TS 20914 Table A.3 checked during development (Results, first subsection)	All six printed quantities and the six individual u(Cref,i) reproduced; worst difference 0.0075 % on U, within the handbook's own rounding
3 · Independent re-implementation	(a) nine models over the full grid of conventions; (b) ANOVA precision estimator; (c) four sources of u(Cref)	check_models.py, check_axes.py (laboratory data); check_prec.py (example data)	(a) 4662 numerical comparisons, largest relative difference 6×10^{-4}, no mismatch > 0.5 %; 1107 declines agree (run of 28 Sep 2026, see §5); (b) and (c) all components agree to $< 10^{-6}$ (output below)
4 · Monte Carlo	Seed reproducibility; agreement with an independent 400 000-draw	check_mc.py	12 of 12 assertions pass; largest coverage-factor difference 0.29 %

	simulation; invariance to the bias point estimate		
5 · User interface	The application driven as a user would: file formats, screens, decision guide, exclusions, downloads, reset	check_ui.py (31 assertions), check_full.py (32 assertions)	63 of 63 pass; no console or page errors

3. Level 2 — worked example of Nordtest TR 537 edition 4 (nordtest_val.py)

The script enters the handbook's ammonium data (six proficiency-testing rounds, control-chart standard deviation 1.67 %) into the application and reads back every intermediate quantity. "HANDBOOK" is the value printed in the handbook; differences are within the handbook's rounding to two decimals.

```
Nordtest TR 537 ed.4 — Appendix 4 / Appendix 9 ammonium example

quantity                  TOOL HANDBOOK      diff
u_Rw %                  1.6700     1.67   +0.0000
RMS_bias %              2.2620     2.26   +0.0020
u(C_ref) %              1.5201     1.52   +0.0001
u(bias) %               2.7253     2.73   -0.0047
u_c %                   3.1963     3.20   -0.0037
U % (k=2)               6.3925     6.40   -0.0075

round bias %: [2.4691, 2.7397, 1.8939, 1.4286, 1.8182, 2.8571]
u(Cref_i) %: [1.7961, 1.1667, 1.4142, 1.6903, 1.1667, 1.8865]  (handbook: 1.80 1.17 1.41 1.69 1.17
1.89)

Errors: none

>>> TOOL REPRODUCES THE PUBLISHED EXAMPLE
```

The handbook's step-4 line "u(Rw) = 3.34/2 = 1.67 %" is a conversion of a ±2s control limit. Read as a general instruction to halve a standard deviation it gives u_c = 2.855 % and U = 5.71 % for this example, which does not match the handbook's own U = 6.40 %; the worked example therefore discriminates between the two readings (Results, first subsection).

4. Level 3b and 3c — precision estimator and u(Cref) sources (check_prec.py)

ANOVA: the application's one-way random-effects decomposition (day as grouping factor, unbalanced estimator of ISO 5725-2, negative between-day component set to zero) against an independent Python implementation on the albumin data built into the application (two levels, 615 and 614 results). u(Cref): the four selectable sources on a traceable measurand (albumin, eight transferred rounds) and on a consensus measurand (alanine aminotransferase, twelve rounds).

```
ANOVA — variance components, tool vs independent re-implementation
  PASS  Level 1      sr     tool 0.654406  python 0.654406
  PASS  Level 1      sbd    tool 0.553962  python 0.553962
  PASS  Level 1      sTot   tool 0.857392  python 0.857392
  PASS  Level 1      degrees of freedom 131/482   (131, 482)
  PASS  Level 1      n0 (unbalanced estimator)  tool 1.269924  python 1.269924
  PASS  Level 2      sr     tool 0.722293  python 0.722293
  PASS  Level 2      sbd    tool 1.074487  python 1.074487
  PASS  Level 2      sTot   tool 1.294693  python 1.294693
  PASS  Level 2      degrees of freedom 130/482   (130, 482)
  PASS  Level 2      n0 (unbalanced estimator)  tool 1.267875  python 1.267875
  PASS  selecting the ANOVA axis feeds s_total into the models   Level 1: tool 2.697663 vs python
2.697663
  PASS  the two precision sources really differ   2.6977 vs 2.7048 %

u(C_ref) source axis — tool vs independent re-implementation
  PASS  example rounds read back from the table   8
  PASS  u(Cref) source nordtest    tool 0.687500  python 0.687500
  PASS  u(Cref) source iso13528    tool 0.687500  python 0.687500
  PASS  u(Cref) source reported    tool 0.687500  python 0.687500
```

```
  PASS  u(Cref) source reportedU   tool 0.687500  python 0.687500
  PASS  consensus rounds, u(Cref) nordtest    tool 0.539557  python 0.539557
  PASS  consensus rounds, u(Cref) iso13528    tool 0.674446  python 0.674446
  PASS  consensus rounds, u(Cref) reported    tool 0.639750  python 0.639750
  PASS  consensus rounds, u(Cref) reportedU   tool 0.319875  python 0.319875
  PASS  the four sources really do differ on a consensus target   {'nordtest': 0.5396, 'iso13528':
0.6744, 'reported': 0.6397, 'reportedU': 0.3199}
  PASS  ISO 13528 is exactly 1.25 x the classical form
  PASS  the expanded reading is exactly half the standard one
  PASS  all albumin rounds are traceable, so the axis cannot move u(Cref)   True -> [0.6875, 0.6875,
0.6875, 0.6875]
  PASS  no page errors

0 failed
```

5. Level 3a — the nine models over the grid of conventions (check_models.py, check_axes.py)

These two scripts need the laboratory's own result files (data/eqa-final.csv, data/iqc-summary.csv and the result tables the application exports for the 60 measurand-levels). The data are not part of the public record because they belong to the laboratory and to Labquality Oy; they are available from the author on reasonable request and the scripts run unchanged on them (the column layout is documented in data/README.md). The outputs below are those recorded when the scripts were run on the study data on 28 September 2026, the run reported in the manuscript.

Script	Grid	Comparisons	Largest relative difference	Mismatches > 0.5 %	Declines
check_models.py	k = 2, RSSu; 9 models × 60 measurand-levels	540	2.6×10^{-4}	0	—
check_axes.py	2 analytic coverage rules (k = 2, Welch–Satterthwaite) × 5 bias treatments (RSSu, RSSU, SUMU, SUMU_MAX, Ue) × 9 models × 60 measurand-levels = 5400 cells	4662 (5400 minus the 738 cells in which the Synek condition $\|\Delta\|/u_c > 0.8$ does not hold)	6.1×10^{-4}	0	1107 (738 above plus 369 in the Monte Carlo arm), declined by both implementations

How the comparison is built: for each measurand-level the application's exported budget (u_Rw, u_cal, the round biases, u(Cref), s_R, p) is recomputed in Python from the same primitives under each convention, and the expanded uncertainty is compared with the value the application prints. A cell counts as a mismatch when the relative difference exceeds 0.5 %; the largest difference observed, 6×10^{-4}, comes from the application's rounding of printed intermediate values. The grid includes the models on which the bias treatment has no effect, so the cells are not all distinct values; they are the combinations a user can select.

6. Level 4 — Monte Carlo coverage (check_mc.py)

Three properties are checked because a stochastic result cannot be matched digit for digit: with a seed the same evaluation returns exactly the same interval and a different seed a different one; an independent Monte Carlo written with numpy (different generator, 400 000 draws) reaches the same coverage factor within 0.3 %; and doubling the bias point estimate leaves the coverage factor unchanged, which is the direct test that the point estimate is applied after the draw and not drawn.

```
  PASS  same seed reproduces the interval exactly   {'iso': 5.360123, 'nordtest_mean': 9.38843,
'eurolab': 11.622927, 'cofrac': 8.487279}
  PASS  a different seed gives a different interval   (9.38843, 9.416827)
  PASS  unseeded runs differ, as they should
```

```
Monte Carlo half-width, tool vs independent implementation (relative difference)
  PASS  iso            k_tool 1.9628  k_python 1.9635  (0.04 %)
  PASS  nordtest_mean  k_tool 2.0019  k_python 2.0079  (0.29 %)
  PASS  nordtest_rms   k_tool 1.9847  k_python 1.9877  (0.15 %)
  PASS  eurolab        k_tool 2.1596  k_python 2.1635  (0.18 %)
  PASS  cofrac         k_tool 2.1631  k_python 2.1679  (0.22 %)
  PASS  iso21748       k_tool 1.9588  k_python 1.9641  (0.27 %)
  PASS  with large degrees of freedom the draw collapses onto 1.96   1.9628
  PASS  doubling the bias point estimate leaves the coverage factor untouched   [2.001936, 2.001936]
  PASS  no page errors

0 failed
```

7. Level 5 — the application driven as a user (check_ui.py, check_full.py)

check_ui.py follows one complete journey on the built-in example (load, define, rounds, result, method laboratory, Levey–Jennings exclusion, record, report set) and asserts what each screen says; it is the check that found the scheme-type side effect described in Methods. check_full.py adds the cases a journey does not meet: a Turkish middleware export with semicolons, decimal commas and day-first dates on two analysers; three control lots on one level; a summary export without individual results; an unreadable file; every screen; all 48 answer combinations of the decision guide; every download; and the full reset. Fixture files are generated by the script itself with a fixed seed.

check_ui.py

```
  PASS  decision guide continues to the 3-step screen, not the axis panel   ['v-simple']
  PASS  report header carries the analyte typed in step 0   Albumin
  PASS  report header carries the sample typed in step 0   Serum/Plasma
  PASS  report header carries the instrument typed in step 0   c703
  PASS  report shows no dash for the measurand
  PASS  EQA scheme type derived automatically as traceable for albumin   ['accuracy', 'auto']
  PASS  result banner states where the scheme type came from
  PASS  guide and result gate do not contradict for a traceable target
  PASS  guide and result gate do not contradict for a consensus target   same class, different
wording: nordtest_sem: sens/warn
  PASS  SEM variant flagged: albumin mean bias is significant   The mean bias is significant (t =
3.90 > t0.975 = 2.36), so
  PASS  result screen draws the model-comparison chart   2
  PASS  chart carries EFLM desirable and minimum separately   ['EFLM desirable=2.50', 'EFLM
minimum=3.75']
  PASS  caption counts models against the limit
  PASS  the record uses the limit chosen in step 3 (EFLM minimum)   {'mu': 3.75, 'src': 'EFLM
minimum', 'inReport': True}
  PASS  the printed report states that limit (EFLM minimum)
  PASS  the record uses the limit chosen in step 3 (RCPA allowable)   {'mu': 8, 'src': 'RCPA
allowable', 'inReport': True}
  PASS  the printed report states that limit (RCPA allowable)
  PASS  choosing no limit prints no verdict rather than a stale one
  PASS  CRM result offers the two routes, with the correction route enabled when significant
{'route': 'block', 'corr': True}
  PASS  a corrected bias enters the ISO budget as a third component   (5.49, {'U': 5.5342, 'rec':
5.5342, 'ub': 0.349, 'field': '0.698'}, 5.5342)
  PASS  an insignificant bias is refused for correction
  PASS  the CRM can be used as a trueness reference (one Xref round)   {'type': 'Xref', 'X': '141.8',
'res': '143.4', 'um': '0.45'}
  PASS  the duplicate multi-round table is gone
  PASS  a single round can be sent to the bias series with its target type and peer statistics
{'n': 9, 'type': 'Xt', 'sd': '1.3', 'np': '63'}
  PASS  a round without peer SD or n is accepted but flagged as optimistic   Added to the bias series
as a Xt round. The series now holds 1 round, mean bias
  PASS  Levey-Jennings charts appear without pressing a button   2
  PASS  marking points alone changes nothing   ('2.70', '2.70')
  PASS  Apply & recalculate refreshes the Estimate MU result   ('2.70', '2.57')
  PASS  saving an analyte adds it to the set   1
  PASS  starting the next analyte keeps the set and clears the data   [1, True, '']
  PASS  no console or page errors during the whole journey
```

```
31 checks, 0 failed
```

check_full.py

```
1 · Turkish headers, decimal commas, day-first dates, two analysers
  PASS  the file is read at all   600
  PASS  raw mode detected   raw
  PASS  both analysers found   ['All analysers (combined)', 'Cihaz 1', 'Cihaz 2']
  PASS  decimal commas parsed as decimals, not thousands   [30.25, 50.39, 30.51]
  PASS  dates read day-first (January, not May)   [0, 0, 0]
  PASS  two levels resolved from the Turkish level column   {'levels': ['Level 1', 'Level 2'], 'n':
[1, 1], 'rw': [2.66, 2.7]}
  PASS  u_Rw lands near the 2.7 % the data was built with   [2.66, 2.7]
  PASS  §6.4 device pooling adds a between-device component   {'pooled': True, 'K': 2, 'uB': 0.0156,
'rw': 2.668}

2 · three reagent lots on one level (ISO/TS 20914 Formula 2 pooling)
  PASS  the three lots are pooled into one level   {'levels': 1, 'lots': 3, 'df': 267, 'rw': 2.712}
  PASS  degrees of freedom add up across lots   267

3 · a summary export with no individual measurements
  PASS  summary mode recognised   summary
  PASS  Levey-Jennings declines politely without raw points   Levey-Jennings / outliers need raw per-
measurement data (summary mode

4 · a file with nothing recognisable in it
  PASS  an unreadable file does not crash the tool   {'status': 'Could not recognise the column
headers automatically — opening the column mapper so you can point at them.', 'mapcard': '',
'crashed': Fal
  PASS  it says something, or offers the column mapper   {'status': 'Could not recognise the column
headers automatically — opening the column mapper so you can point at them.', 'mapcard': '',
'crashed': Fal

5 · screens the routine journey does not open
  PASS  core     opens without error
  PASS  pool     opens without error
  PASS  about    opens without error
  PASS  guide    opens without error
  PASS  methods  opens without error
  PASS  eqa      opens without error
  PASS  bias     opens without error
  PASS  report   opens without error
  PASS  qc       opens without error
  PASS  Formulas & sources names every model family

6 · decision guide, every combination of answers
  PASS  every answer combination judges all nine models
  PASS  with no trueness reference only the ISO model stays available

7 · the files the tool hands out
  PASS  blank IQC template downloads
  PASS  blank EQA template downloads
  PASS  method-comparison appendix exports as CSV
  PASS  printable report downloads

8 · full reset
  PASS  reset clears data, series, set and the corrected-bias term   {'iqc': True, 'set': 0, 'rows':
0, 'ub': None, 'analyte': '', 'out': 0}
  PASS  no console or page errors in the whole pass

0 checks failed, 0 notes
```

8. Defects found by these checks

Three defects in the application were found by the checks and corrected before the data were analysed (Methods, "Verification"): the Monte Carlo arm drew the bias point estimate as a random component (mean-bias form on albumin 9.40 % → 10.46 %); a routine re-derived the scheme type while the applicability screen was judging by it; and a specification limit chosen on one screen did not reach the printed record. Two defects in the

released check scripts themselves were found when this report was regenerated on a clean machine and are corrected in the repository: check_prec.py read the example file with a fixed comma delimiter and compared against a different example file than the one built into the application, and check_full.py expected its fixture files at a fixed path instead of generating them. Neither affected the application or any result in the manuscript; both prevented a reader from re-running the checks without the author's machine, which is the point of distributing them.

9. How to reproduce this file

```
git clone https://github.com/mfkilinckaya-svg/VerifyMyMU   # or unpack the Zenodo archive
cd VerifyMyMU
python3 -m venv venv && . venv/bin/activate
pip install numpy scipy playwright && playwright install chromium
python verification/nordtest_val.py
python verification/check_prec.py
python verification/check_mc.py
python verification/check_ui.py
python verification/check_full.py
# with the laboratory data in data/:
python verification/check_models.py && python verification/check_axes.py
```

Each script prints PASS/FAIL per assertion and a one-line total; a non-zero exit status or any FAIL line means the file in hand is not the one verified here.